

Shantanu Shastri

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EDUCATION

Carnegie Mellon University	December 2026
Master of Information Systems Management	GPA: 3.85
Manipal Institute of Technology	August 2022
Bachelor of Electronics and Communications Engineering	GPA: 3.99
<i>Relevant Coursework:</i> Network Security, Distributed Systems, Data Structures, Algorithms, System Design	

SKILLS

Languages: C, C++, Golang, Rust, Python, Java, JavaScript, SQL, Bash/Shell
Networking & Security: HTTP, TLS, TCP/IP, DNS, Identity Access Management, Network Security
Backend Architecture: Unix/Linux, Distributed Systems, REST APIs, System Design, Microservices
Infrastructure & DevOps: Azure, Docker, Kubernetes, Apache Kafka, Zookeeper, CI/CD, Live Site Monitoring

EXPERIENCE

BlackRock	January 2022 – August 2026
<i>AI Engineer Internship</i>	June 2026 – August 2026

- **Secure Distributed Systems:** Built a secure, large-scale distributed backend processing 500+ post-trade exceptions/week using Golang and C++, ensuring strict network security over TCP/IP and HTTP protocols.
- **AI for Security & Productivity:** Leveraged AI to boost engineering productivity by building a secure generative AI platform, backed by highly diagnosable and actively monitored RESTful APIs handling 2K+ RPS.
- **Telemetry & Data Drift:** Created offline-online evaluation frameworks leveraging telemetry to track data drift and anomalies, safely processing 3.5M records and \$14T in cash with extreme reliability.

<i>Software Developer 2 - Associate</i>	January 2025 – July 2025
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- **System Design & Scale:** Architected a Java Spring Boot secure data framework and TTL Azure microservices, automating 1.5M+ client touchpoints with well-defined interfaces and robust access controls.
- **Live Site Reliability:** Engineered a distributed read-through NoSQL cache with LRU eviction for Live Site reliability, scaling concurrent requests and cutting response times from 450ms to 150ms.

<i>Software Developer 1 - Analyst</i>	July 2022 – December 2024
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- **Unix/Linux Infrastructure:** Resolved singleton bottlenecks by engineering a sharded Unix/Linux server environment with Zookeeper, boosting system throughput to 10K+ RPS for mission-critical services.
- **Code Reviews & Testing:** Drove modern engineering practices through rigorous design and code reviews, building Azure CI/CD pipelines with comprehensive integration tests to improve test coverage.
- **Database Optimization:** Took end-to-end ownership of database optimization, refactoring PostgreSQL schemas on tables exceeding 50M+ rows to achieve a 37% reduction in query latency.

Shunya	June 2020 – September 2020
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- Data Science Intern*
- **Linux Workflows:** Engineered a GPU-accelerated PyTorch OpenCV workflow on a Linux environment, successfully automating evaluations and reducing manual grading time by 90%.
 - **Data-Driven Insights:** Addressed learning gaps by training clustering models on multidimensional data to dynamically recommend targeted practice questions, utilizing data-driven approaches.

PROJECTS

Zulip Open-Source Engine Extensions

- **Open-Source Contribution:** Contributed to open-source by building a high-throughput message engine using Node.js and Redis over TCP/IP, securely processing live data streams with zero message loss.

High-Performance Modular Booking Architecture

- **Distributed Architecture:** Engineered a secure, decoupled microservices architecture using Apache Kafka and TLS/HTTP protocols, handling bursts of 100,000+ real-time requests with Docker deployments.